

Distributed Multi-agent System for Time-varying Quadratic Programming with Application to Target Encirclement of Multi-robot System

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Abstract—In this paper, a unified distributed multi-agent system is proposed to minimize time-varying quadratic function under time-varying coupled equality constraint, which is further applied to target encirclement of multi-robot system. The time-varying quadratic programming problem under consideration is a generalized version of some found in the literature. The proposed unified distributed multi-agent system is effective in both of the following cases of time-varying quadratic programming: one with nonidentical time-invariant Hessian and constraint matrices, and the other with identical time-varying Hessian and constraint matrices. The convergence of the time-varying states can be guaranteed if the graph of communication network is connected and undirected, subject to certain mild conditions. Additionally, simulations show that the suggested approach is effective in resolving the target encirclement problem of multi-robot systems.

Index Terms—Time-varying quadratic programming, multi-robot system, distributed multi-agent system, target encirclement.

I. INTRODUCTION

In recent decades, optimization modeling and algorithms have been intensively investigated due to their widespread applications in engineering, such as robot navigation [1], formation control [2]–[4], machine learning [5], resource allocation [6]–[8] and congestion control [9]. These practical problems are often modeled as optimization problems involving convex or non-convex objective functions with constraints.

Among the optimization methods, dynamic optimization models based on neurodynamic systems or multi-agent systems are extensively studied recently [10]–[14]. A one-layer recurrent neural network presented in [15] provides a useful method for optimization with pseudoconvex cost functions, linear equality constraints, and bound constraint conditions. Moreover, a single-layer recurrent neural network, as presented in [16], is used to resolve pseudoconvex optimization problems subject to both equality and inequality constraints. However, this network does not require any penalty parameter. To solve the L_1 -minimization problem, [17] provides several algorithms with considering the continuous-time projection neural network for sparse signal reconstruction. Additionally, in recent years, there has been a lot of discussion about optimization

algorithmss based on continuous-time dynamic systems with fixed-time or finite-time convergence. For example, the fixed-time algorithms to solve linear and quadratic programming problems are proposed in [18] and [19] respectively.

However, It is important to emphasize that the cost function and constraint in practical optimization problem may be explicitly time dependent, i.e., time-varying optimization problem is more desirable in real engineering applications. For this problem, the optimal solution is a function with respect to time. In early research [20], it takes a snapshot of the time-dependent optimization issue at a given moment in time, during which it is presumed to be time-invariant, and the time-varying problem is subsequently solved using traditional time-invariant optimization algorithms. Although this approach can address a few time-varying optimization problems, it is likely to introduce substantial tracking error if the solution trajectory evolves swiftly during the sampling periods. Therefore, in order to overcome this drawback, some innovative approaches have emerged. For instance, a class of discrete-time prediction-correction method is developed in [21] for minimizing time-varying convex objective functions. With a particular focus on time-varying constraint optimization problems, a discrete-time prediction-correction method is presented without computing the inverse of Hessian matrix in objective function [22], which exhibits augmented asymptotic tracking. It is noted that the algorithms discussed above operate in discrete time. However, as many real-world systems operate in continuous time, numerous continuous-time optimization techniques have been put forth. Specifically, a prediction-correction scheme, which has a foundation for the interior-point method, is created in [23] to minimize time-varying function subject to time-varying inequality and equality constraints. Moreover, this scheme achieves an asymptotically vanishing tracking error between the solution of the designed approach and the optimal trajectory. Based on the initial zeroing neural network method [24], an innovative zeroing neural network is designed in [25] with finite-time convergence rate and noise-tolerant property, which makes it applicable to solve nonlinear optimization problem subject to time-varying equality constraint.

several researchers have focused their attention on distributed optimization algorithms recently. In the context of such algorithms, each agent in the system is only accessible to partial (and possibly private) information, then all agents aim to cooperatively solve optimization problems by local information exchange [26], [27]. When it comes to the optimization problems involving both global and local equality

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and inequality constraints as well as hybrid constraints, [7] and [28] propose some distributed discrete algorithms to solve such problems. In contrast to the distributed fixed-time algorithm in [29], which is only useful for the convex programming with equality constraint, the algorithm in [30] is applicable for the constraint fixed-time problems. For addressing time-varying optimization problem subject to equality constraint, [31] offers a distributed algorithm. A distributed continuous-time consensus-based algorithm derived from neurodynamic system is provided in [32] to solve time-varying problems with inequality constraint. This algorithm gets all agents to agree on the optimal solution by using a log-barrier and signum penalty function. In [14], two distributed continuous-time approaches are presented for addressing time-dependent optimal resource allocation problem, taking time-varying objective function and equality constraint into consideration. These algorithms are applicable for nonidentical time-invariant and identical time-varying Hessian matrices, respectively. Furthermore, a fixed-time algorithm in distributed manner is designed in [33] for minimizing time-varying convex functions, allowing all agents to reach consensus within a predetermined time period. In contrast to the technique given in [14], it is not required that all local objective functions have identical Hessian matrix or follow a predefined format, but it does work only if there is no constraint to the optimization problem. In [8], a distributed method based on continuous-time dynamical system is presented to solve time-varying quadratic programming problems in which Hessian matrix of local objective function is in diagonal form but the weight coefficient in equality constraint is a scalar function. However, in the work of this paper, we mainly focus on two cases of time-varying quadratic programming problems: one is that the Hessian and constraint matrices for the optimization problem are time-invariant and different, and the other one is that they are time varying and identical. However, different from the one-dimensional time-varying Hessians in [14], the Hessians of local objective functions are time-varying matrices in this paper. Moreover, the weight coefficients and total resources in the coupled equality constraint are constants in [14] but vectors in this paper. More importantly, here we propose a distributed continuous-time multi-agent system for addressing time-varying optimization problem that unifies the two aforementioned cases into one framework.

The major contributions of this paper are presented as follows. First, for the time-varying quadratic optimization problem subject to coupled equality constraint, we consider two cases in this paper: one with nonidentical time-invariant Hessian and constraint matrices, and the other with identical time-varying Hessian and constraint matrices. Both cases are also considered in [14]. Different from [14] where two cases are analyzed and resolved separately, this paper proposes a unified distributed continuous-time multi-agent system that works for the two scenarios. In addition, unlike the Hessians in [34] and the total resource in [14] are constants, the Hessians are matrices that may be time-varying and the total resource is a time-varying vector in this paper. This means that the time-varying optimization problem investigated in this paper is more complicated and more general, so the proposed multi-

agent system is appropriate for a wide range of scenarios. The differences mentioned above lead to difficulties in theoretical analysis. Second, for the purpose of estimating global information obtained by the prediction-correction method in a distributed manner [35], a distributed average tracking approach is combined into the proposed continuous-time multi-agent system in this paper. The proposed method ensures that each agent follows its ideal trajectory with minimal deviation. Third, we show that the suggested approach is suitable for the optimal time-varying target encirclement of multi-robot system. That is, the proposed multi-agent system can encircle a moving target cooperatively.

The remaining parts of the paper are structured as shown below. In Section II, we present some notations, preliminaries on graph theory and some results on consensus algorithms. Section III describes the optimization problem and multi-agent systems in details. The convergence analysis of the presented distributed multi-agent system is detailed in Section IV, demonstrating its capability to track optimal trajectories in two scenarios. In section V, a practical application of the proposed method in target encirclement of multi-robot system is demonstrated firstly, then a numerical simulation and experimental results are provided. Finally, the conclusion of this paper is presented in section VI.

II. PRELIMINARIES

A. Notations

In this paper, we use the symbol $\mathbb{R}$ to represent the set of real numbers, $\mathbb{R}_+$ for non-negative real numbers, and $\mathbb{R}_{++}$ for positive real numbers. The symbol $\mathbb{R}^n$ is used for the set of n -dimensional real vectors. The p -norm of vector $x \in \mathbb{R}^n$ or matrix A is denoted by $\|x\|_p$ or $\|A\|_p$, where p takes the value of 1, 2 and ∞ . Let $\mathbb{R}^{m \times n}$ stand for the collection of all $m \times n$ real matrices. The transpose of matrix A is indicated by A^T . Let I_m denote a m -dimensional identity matrix. If two matrices A and B satisfy $A \succeq B$, it refers to $A - B$ is positive semi-definite. Matrix A is vectorized when it is transformed into a column vector by stacking its columns, as shown by the notation $\text{vec}(A)$. Conversely, for a vector α , $\text{mat}(\alpha)$ denotes a matrix generated from α in columns with appropriate dimension. Let $\text{sign}(x)$ stand for the signum function for $x \in \mathbb{R}$. This means that, $\text{sign}(x) = 1$ if $x > 0$; $\text{sign}(x) = 0$ if $x = 0$; $\text{sign}(x) = -1$ otherwise. Furthermore, we write $\text{sign}(x) = (\text{sign}(x_1), \text{sign}(x_2), \dots, \text{sign}(x_n))^T$ for a vector $x = (x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n$. Considering a function $f(x, t) : \mathbb{R}_m \times \mathbb{R}_+ \rightarrow \mathbb{R}$, the gradient and Hessian matrix of $f(x, t)$ are denoted by the notation $\nabla_x f(x, t) \in \mathbb{R}^m$ and $H(x, t) \in \mathbb{R}^{m \times m}$, respectively. Let $\nabla_{xt} f(x, t) \in \mathbb{R}^n$ denote the partial derivative of $\nabla_x f(x, t)$ with respect to t .

B. Graph Theory

In this paper, we take into account an undirected graph, denoted by $\mathcal{G} = (\mathcal{V}, \mathcal{E}, A)$, which is used to define the communication network of multi-agent system. The graph of the network is composed of a set of vertices $\mathcal{V} = \{v_1, v_2, \dots, v_m\}$, a set of edges $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$, and a matrix of adjacency relations between vertices $A = \{a_{pq}\}_{m \times m}$ with $a_{pq} = 1$

if $(v_p, v_q) \in \mathcal{E}$ and $a_{pq} = 0$ otherwise. When two nodes (v_p, v_q) are connected by an edge e_{pq} , it indicates that the two nodes v_p and v_q can communicate with one another. Let $N_p = \{q \in \mathcal{V} : e_{pq} \in \mathcal{E}\}$ represent the collection of nodes adjacent to node p in the graph. Throughout this paper, we ignore self-loops in the graph by assuming that $a_{pp} = 0$. The degree of node v_p is typically defined as $d_p = \sum_{q=1}^m a_{pq}$.

The Laplacian matrix, which is generally defined as $L = D - A$ with $D = \text{diag}\{d_1, d_2, \dots, d_m\}$, plays an important role in graph theory to help grasp some properties of the graph. Specifically, it is used to encode the connectivity and weight information of the underlying graph structure. A path from v_p to v_q in the graph is denoted by the set of edges $(v_p, v_{p1}), (v_{p1}, v_{p2}), \dots, (v_{pk}, v_q)$, where each edge connects two different nodes in the graph. Connectivity in graph $\mathcal{G}$ means there is always a path to get from one node to another. Moreover, if graph $\mathcal{G}$ is connected and undirected, the eigenvalues satisfy $\lambda_1(L) = 0 < \lambda_2(L) \leq \dots \leq \lambda_m(L)$ with $\lambda_k(L)$ denoting the k th smallest eigenvalue of the Laplacian matrix.

For later analysis, we make the assumption that the interaction graph of the multi-agent system under consideration satisfies the following condition.

Assumption 1. The interaction graph of the multi-agent system is both connected and undirected.

C. Consensus Algorithms

The consensus algorithms of multi-agent systems are crucial in the design and evaluation of distributed optimization. This type of algorithms enables agents in a distributed system to achieve agreement in terms of a predefined optimizing criterion. It is used to provide a basis for distributed coordination, communication, and optimization among individual agents within a system. Next, we introduce two consensus algorithms, which are exploited for subsequent analyses.

We introduce the following dynamical system for agent i

$$\begin{aligned} \dot{z}_i(t) &= \gamma \sum_{j \in N_i} \text{sign}(o_j(t) - o_i(t)), \\ o_i(t) &= z_i(t) + \zeta_i(t), \end{aligned} \quad (1)$$

in which $\gamma > 0$ is a constant, $o_i \in \mathbb{R}^n$ is the output vector of the system, $z_i \in \mathbb{R}^n$ satisfies $\sum_{i=1}^m z_i(0) = 0$, $\zeta_i \in \mathbb{R}^n$ is time-varying reference signal of agent i , and $\dot{\zeta}_i(t)$ is bounded, i.e., there exists $\bar{\zeta} > 0$ satisfying $\sup_{t \geq 0} \|\dot{\zeta}_i(t)\|_\infty \leq \bar{\zeta}$.

Lemma 1. [35] Under Assumption 1, the output vector $o_i(t)$ of system (1) will reach consensus within a finite time if the design parameter γ satisfies $\gamma > \bar{\zeta}$; i.e., there exists a finite time $T > 0$ estimated by $T \leq (\gamma - \bar{\zeta})/2 \sum_{i=1}^m \sum_{j \in N_i} \|o_i(0) - o_j(0)\|_\infty$, and we have $o_i(t) = (\sum_{i=1}^m \zeta_i(t))/m$ for $t \geq T$.

The second considered system for the i th agent is described as the following continuous-time dynamic system

$$\dot{z}_i(t) = -\Xi(t)z_i(t) + v_i(t) - \alpha \sum_{j \in N_i} \text{sign}(z_i(t) - z_j(t)), \quad (2)$$

where $\Xi(t)$ is a time-varying coefficient matrix, α is a positive constant, $z_i \in \mathbb{R}^n$ is the state vector of the i th agent,

and $v_i(t) \in \mathbb{R}^n$ is a time-varying bounded vector function satisfying $\sup_{t \geq 0} \|v_i(t)\|_\infty < \kappa$.

Lemma 2. Under Assumption 1, the state vector $z_i(t)$ of multi-agent system (2) will reach consensus if the design parameter α satisfies $\alpha \geq \kappa \sqrt{2nm/\lambda_2(L)}$ with $\lambda_2(L)$ being the second smallest eigenvalue of matrix L , and the coefficient matrix $\Xi(t)$ satisfies $\Xi(t) + \Xi^T(t) \succeq 2\tau I_n$ with τ being a positive constant.

Proof: See Appendix A. ■

Remark 1. Unlike the case where $\Xi(t) \in \mathbb{R}_{++}$ is a constant in [14], $\Xi(t)$ in the dynamical system (2) is a $n \times n$ -dimensional time-varying matrix.

III. PROBLEM FORMULATION AND ALGORITHM DESIGN

In this paper, our emphasis is on time-varying quadratic programming with coupled equality constraint, generally characterized as

$$\begin{aligned} \min_{x(t)} \quad & \sum_{j=1}^m f_j(x_j(t), t), \\ \text{s.t.} \quad & \sum_{j=1}^m D_j(t)x_j(t) = b(t), \end{aligned} \quad (3)$$

in which $x(t) = (x_1^T(t), x_2^T(t), \dots, x_m^T(t))^T$ with $x_j(t) \in \mathbb{R}^n$ being the state of the j th agent ($x_j(t)$ is often written as x_j for simplicity), $D_j(t) \in \mathbb{R}^{l \times n}$ is time-varying coefficient matrix for x_j which needs to be full row-rank for all $t \geq 0$, $b(t) \in \mathbb{R}^l$ is also time-varying, and $f_j(x_j, t)$ is time-varying objective function of agent j with the following quadratic form

$$f_j(x_j, t) = \frac{1}{2} x_j^T Q_j(t) x_j + c_j^T(t) x_j + d_j(t), \quad (4)$$

where $Q_j(t) \in \mathbb{R}^{n \times n}$, $c_j(t) \in \mathbb{R}^n$ and $d_j(t) \in \mathbb{R}$.

Remark 2. Time-varying quadratic programming subject to coupled equality constraint (3) has been investigated in [14], [34]. However, the Hessian $Q_j(t) \in \mathbb{R}_{++}$ in [34] and $b(t) \in \mathbb{R}$ in [14] are both constants, $Q_j(t) \in \mathbb{R}_{++}$ in [14] is one-dimensional. In addition, the weight coefficient $D_j(t) = 1$ in [14] and [34]. The disparities indicate that applying the algorithms from [14], [34] directly to problem (3) is not feasible and, moreover, poses challenges for theoretical analysis.

Next, for further analysis, we need to make the following assumptions.

Assumption 2. The cost function $f_j(x_j, t)$ in (4) is uniformly strongly convex for $t \geq 0$; i.e., $Q_j(t) \succeq hI_n$ for some $h > 0$. Moreover, the elements of $Q_j(t)$, $c_j(t)$, $\dot{Q}_j(t)$ and $\dot{c}_j(t)$ are uniformly bounded ($j \in \{1, 2, \dots, m\}$).

Assumption 3. The Slater's condition holds for the optimization problem (3); i.e., there exists at least one $x(t) \in \mathbb{R}^{mn}$ to satisfy the equality constraint in (3) for $t \geq 0$. Moreover, the elements of $D_j(t)$ and $\dot{D}_j(t)$ are uniformly bounded ($j \in \{1, 2, \dots, m\}$).

Next, we define the Lagrange function of (3) as follows

$$\Psi(x, \mu, t) = \sum_{j=1}^m f_j(x_j, t) + \mu^T(t) \left(\sum_{j=1}^m D_j(t)x_j - b(t) \right), \quad (5)$$

in which $\mu(t) \in \mathbb{R}^l$ is the Lagrange multiplier, which is often written as μ for simplicity. If we divide the vector $b(t)$ such that $\sum_{j=1}^m b_j(t) = b(t)$, the Lagrange function may be alternatively expressed as

$$\Psi(x, \mu, t) = \sum_{j=1}^m f_j(x_j, t) + \mu^T \left(\sum_{j=1}^m (D_j(t)x_j - b_j(t)) \right).$$

Then the Hessian matrix of the Lagrange function is

$$\tilde{Q}(t) = \begin{pmatrix} Q_1(t) & O & \cdots & O & D_1^T(t) \\ O & Q_2(t) & \cdots & O & D_2^T(t) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & Q_m(t) & D_m^T(t) \\ D_1(t) & D_2(t) & \cdots & D_m(t) & O \end{pmatrix}, \quad (6)$$

where O is zero matrix with appropriate dimension.

As per the prediction-correction method [14], [23], if Assumptions 2 and 3 hold, the vector $(x^T, \mu^T)^T$ satisfying

$$\tilde{Q}(t) \begin{pmatrix} \dot{x}(t) \\ \dot{\mu}(t) \end{pmatrix} = -\sigma \begin{pmatrix} \nabla_x \Psi(x, \mu, t) \\ \nabla_\mu \Psi(x, \mu, t) \end{pmatrix} - \begin{pmatrix} \nabla_{xt} \Psi(x, \mu, t) \\ \nabla_{\mu t} \Psi(x, \mu, t) \end{pmatrix}, \quad (7)$$

is globally exponentially convergent to the optimal trajectories $((x^*)^T(t), (\mu^*)^T(t))^T$ for $\sigma > 0$.

Combining with (5) and (6), the dynamical system (7) for agent i may be rendered as

$$Q_i(t)\dot{x}_i + D_i^T(t)\dot{\mu} = -\sigma \nabla_{x_i} f_i(x_i, t) - \sigma D_i^T(t)\mu - \nabla_{x_i t} f_i(x_i, t) - \dot{D}_i^T(t)\mu, \quad (8a)$$

$$\sum_{i=1}^m D_i(t)\dot{x}_i = -\sigma \left(\sum_{i=1}^m D_i(t)x_i - b(t) \right) - \left(\sum_{i=1}^m \dot{D}_i(t)x_i - \dot{b}(t) \right), \quad (8b)$$

where $\nabla_{x_i} f_i(x_i, t) = Q_i(t)x_i + c_i(t)$ and $\nabla_{x_i t} f_i(x_i, t) = \dot{Q}_i(t)x_i + \dot{c}_i(t)$.

Since $Q_i(t)$ is invertible, from (8a), one gets

$$\dot{x}_i = -Q_i^{-1}(t) \left(D_i^T(t)\dot{\mu} + \sigma \nabla_{x_i} f_i(x_i, t) + \sigma D_i^T(t)\mu + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right). \quad (9)$$

Then multiplying the both sides of (9) by $D_i(t)$ and summing over all agents, we have

$$\sum_{i=1}^m D_i(t)\dot{x}_i = - \sum_{i=1}^m D_i(t)Q_i^{-1}(t) \left(D_i^T(t)\dot{\mu} + \sigma \nabla_{x_i} f_i(x_i, t) + \sigma D_i^T(t)\mu + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right). \quad (10)$$

Throughout the whole of this work, we always suppose $\sum_{i=1}^m D_i(t)Q_i^{-1}(t)D_i^T(t)$ to be invertible for all $t \geq 0$, then substituting (10) into (8b), we can get

$$\dot{\mu} = \left(\sum_{i=1}^m D_i(t)Q_i^{-1}(t)D_i^T(t) \right)^{-1} \left[- \sum_{i=1}^m D_i(t)Q_i^{-1}(t) \right.$$

$$\left. \times \left(\sigma \nabla_{x_i} f_i(x_i, t) + \sigma D_i^T(t)\mu + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right) + \sigma \left(\sum_{i=1}^m D_i(t)x_i - b(t) \right) + \left(\sum_{i=1}^m \dot{D}_i(t)x_i - \dot{b}(t) \right) \right]. \quad (11)$$

Let $\Lambda(t) = \sum_{i=1}^m D_i(t)Q_i^{-1}(t)D_i^T(t)$, then (11) becomes

$$\begin{aligned} \dot{\mu} &= \Lambda^{-1}(t) \left[- \sum_{i=1}^m D_i(t)Q_i^{-1}(t) \left(\sigma \nabla_{x_i} f_i(x_i, t) + \sigma D_i^T(t)\mu + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right) \right. \\ &\quad \left. + \sigma \left(\sum_{i=1}^m D_i(t)x_i - b(t) \right) + \left(\sum_{i=1}^m \dot{D}_i(t)x_i - \dot{b}(t) \right) \right] \\ &= \Lambda^{-1}(t) \left[- \sum_{i=1}^m D_i(t)Q_i^{-1}(t) \left(\sigma \nabla_{x_i} f_i(x_i, t) + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right) + \sigma \left(\sum_{i=1}^m D_i(t)x_i - b(t) \right) \right. \\ &\quad \left. + \left(\sum_{i=1}^m \dot{D}_i(t)x_i - \dot{b}(t) \right) \right] - \sigma \mu. \end{aligned} \quad (12)$$

According to the equations in (9) and (12), the centralized algorithm in (8) can be formulated equivalently as

$$\dot{x}_i = -Q_i^{-1}(t) \left(D_i^T(t)\dot{\mu} + \sigma \nabla_{x_i} f_i(x_i, t) + \sigma D_i^T(t)\mu + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right), \quad (13a)$$

$$\begin{aligned} \dot{\mu} &= \Lambda^{-1}(t) \left[- \sum_{i=1}^m D_i(t)Q_i^{-1}(t) \left(\sigma \nabla_{x_i} f_i(x_i, t) + \nabla_{x_i t} f_i(x_i, t) + \dot{D}_i^T(t)\mu \right) + \sigma \left(\sum_{i=1}^m D_i(t)x_i - b(t) \right) \right. \\ &\quad \left. + \left(\sum_{i=1}^m \dot{D}_i(t)x_i - \dot{b}(t) \right) \right] - \sigma \mu. \end{aligned} \quad (13b)$$

By the expression of $\Lambda(t)$, it is computed in a centralized manner. For the purpose of easing up on the restriction to get $\Lambda(t)$ for each agent in a distributed manner, motivated by the method in [8], we offer the following continuous-time distributed algorithm to estimate $\Lambda(t)$ for agent i

$$\dot{\xi}_i(t) = \gamma \sum_{j \in N_i} \text{sign}(z_j - z_i), \quad (14a)$$

$$z_i(t) = \xi_i(t) + \theta_i(t), \quad (14b)$$

in which $\gamma > 0$ is a constant, and $\theta_i(t) = \text{vec}(D_i(t)Q_i^{-1}(t)D_i^T(t))$.

According to the definition of $\theta_i(t)$ and Assumptions 2-3, $\sup_{t \geq 0} \|\theta_i(t)\|_\infty$ is bounded. From Lemma 1, the output vector $z_i(t)$ of multi-agent system (14) can reach consensus within finite time if the design parameter γ satisfies $\gamma > \bar{\theta} \geq \sup_{t \geq 0} \|\theta_i(t)\|_\infty$. On the other words, there exists a finite time $T > 0$ with $T \leq (\gamma - \bar{\theta})/2 \sum_{i=1}^m \sum_{j \in N_i} \|z_i(0) - z_j(0)\|_\infty$ such that $z_i(t) = (\sum_{i=1}^m \theta_i(t))/m$ for $t \geq T$. We define $\Lambda_i(t) = \text{mat}(mz_i(t)) \in \mathbb{R}^{l \times l}$, where $\text{mat}(mz_i(t))$ is a matrix generated from the vector $mz_i(t)$ in columns. Then $\Lambda_i(t) = \Lambda_j(t) = \Lambda(t)$ for any $t \geq T$.

Furthermore, from the centralized algorithm (13), we can derive a distributed multi-agent system with the i th agent's state equations as

$$\begin{aligned} \dot{\hat{x}}_i = & -Q_i^{-1}(t) \left(D_i^T(t) \dot{\hat{\mu}}_i + \sigma \nabla_{x_i} f_i(\hat{x}_i, t) + \sigma D_i^T(t) \hat{\mu}_i \right. \\ & \left. + \nabla_{x_i} f_i(\hat{x}_i, t) + \dot{D}_i^T(t) \hat{\mu}_i \right), \end{aligned} \quad (15a)$$

$$\begin{aligned} \dot{\hat{\mu}}_i = & m \Lambda_i^{-1}(t) \left[-D_i(t) Q_i^{-1}(t) (\sigma \nabla_{x_i} f_i(\hat{x}_i, t) + \nabla_{x_i} f_i(\hat{x}_i, t) \right. \\ & \left. + \dot{D}_i^T(t) \hat{\mu}_i) + \sigma (D_i(t) \hat{x}_i - b_i(t)) + (\dot{D}_i(t) \hat{x}_i - \dot{b}_i(t)) \right] \\ & - \sigma \hat{\mu}_i - \alpha \sum_{j \in N_i} \text{sign}(\hat{\mu}_i - \hat{\mu}_j), \end{aligned} \quad (15b)$$

where $\alpha > 0$, $\hat{x}_i \in \mathbb{R}^n$ and $\hat{\mu}_i \in \mathbb{R}^l$ are the estimators of x_i and μ for agent i respectively, $\Lambda_i(t) = \text{mat}(m z_i(t))$, and $b_i(t) \in \mathbb{R}^l$ satisfies $\sum_{i=1}^m b_i(t) = b(t)$.

Next, we introduce several lemmas that will be beneficial to the convergence analysis of the distributed multi-agent system.

Lemma 3. Define $w_i = Q_i(t)x_i + D_i^T(t)\mu$ and $\hat{w}_i = Q_i(t)\hat{x}_i + D_i^T(t)\hat{\mu}_i$, then we have $\lim_{t \rightarrow \infty} (\hat{w}_i - w_i) = 0$.

Proof: See Appendix B. ■

Lemma 4. For the centralized algorithm (13) and distributed algorithm (15) with (14), there exists a finite time $T > 0$ such that the following equality holds for any $t \geq T$

$$\begin{aligned} & \frac{1}{m} \sum_{i=1}^m \dot{\hat{\mu}}_i - \dot{\mu} \\ = & -\Lambda^{-1}(t) \sum_{i=1}^m (D_i(t) Q_i^{-1}(t) \dot{Q}_i(t) - \dot{D}_i(t)) Q_i^{-1}(t) \\ & \times (\hat{w}_i - w_i) - \sum_{i=1}^m \left\{ \Lambda^{-1}(t) \left[D_i(t) Q_i^{-1}(t) (\dot{D}_i^T(t) \right. \right. \\ & \left. \left. - \dot{Q}_i(t) D_i^T(t)) + \dot{D}_i(t) D_i^T(t) \right] + \frac{\sigma I_l}{m} \right\} (\hat{\mu}_i - \mu). \end{aligned} \quad (16)$$

Proof: See Appendix C. ■

Lemma 5. Assume the Hessian matrix $Q_i(t)$ and constraint matrix $D_i(t)$ are time-invariant and nonidentical; i.e., $Q_i(t) = Q_i$ and $D_i(t) = D_i$ ($i = 1, 2, \dots, m$). The state vector $\hat{\mu}_i$ of the distributed multi-agent system in (15) with $\Lambda_i(t) = \Lambda_i = \sum_{i=1}^m (D_i Q_i^{-1} D_i^T)$ will reach consensus, i.e., $\lim_{t \rightarrow \infty} (\hat{\mu}_i - \hat{\mu}_j) = 0$, if the parameter α satisfies

$$\alpha \geq \kappa \sqrt{\frac{2nm}{\lambda_2(L)}}, \quad (17)$$

with κ satisfying

$$\kappa \geq m \sup_{t \geq 0} \|\Lambda_i^{-1} [D_i Q_i^{-1} (\sigma c_i(t) + \dot{c}_i(t)) + \sigma b_i(t) + \dot{b}_i(t)]\|_{\infty}, \quad (18)$$

for $i = 1, 2, \dots, m$.

Proof: See Appendix D. ■

Lemma 6. Assume the time-varying Hessian matrix $Q_i(t)$ and constraint matrix $D_i(t)$ are identical respectively; i.e., $Q_i(t) = Q(t)$ and $D_i(t) = D(t)$ ($i = 1, 2, \dots, m$). The state vector $\hat{\mu}_i$ of the distributed multi-agent system in (15) with

$\Lambda_i(t) = \Lambda(t) = m(D(t)Q^{-1}(t)D^T(t))$ will reach consensus, i.e., $\lim_{t \rightarrow \infty} (\hat{\mu}_i - \hat{\mu}_j) = 0$, if the parameter σ satisfies

$$\sigma I_l + \frac{1}{2} (\Gamma(t) + \Gamma^T(t)) \succeq \tau I_l, \quad (19)$$

in which $\tau > 0$ is a constant and $\Gamma(t) = m\Lambda^{-1}(t) [(\dot{D}(t) - D(t)Q^{-1}(t)\dot{Q}^T(t))Q^{-1}(t)D^T(t) + D(t)Q^{-1}(t)\dot{D}^T(t)]$, and α satisfies

$$\alpha \geq \kappa \sqrt{\frac{2nm}{\lambda_2(L)}}, \quad (20)$$

with κ satisfying

$$\begin{aligned} \kappa \geq & m \sup_{t \geq 0} \|\Lambda^{-1}(t) (\dot{D}(t) - D(t)Q^{-1}(t)\dot{Q}^T(t))Q^{-1}(t)\|_{\infty} \\ & \times \left(\sup_{t \geq 0} \|c_i(t) + \dot{c}_i(t)/\sigma\|_{\infty} + \|Q(0)\hat{x}_i(0) \right. \\ & \left. + D^T(t)\hat{\mu}_i(0)\|_{\infty} \right) + m \sup_{t \geq 0} \|\Lambda^{-1}(t) [D(t)Q^{-1}(t)(\sigma c_i(t) \\ & + \dot{c}_i(t)) + \sigma b_i(t) + \dot{b}_i(t)]\|_{\infty}, \end{aligned} \quad (21)$$

for $i = 1, 2, \dots, m$.

Proof: See Appendix E. ■

IV. CONVERGENCE ANALYSIS

In this section, we inspect the convergence of the designed distributed time-varying optimization method (15) in addressing problem (3).

A. With Time-invariant and Nonidentical Hessian and Constraint Matrices

Next we turn attention to the following time-varying quadratic programming problem

$$\begin{aligned} \min_{x(t)} \quad & \sum_{i=1}^m \frac{1}{2} x_i^T Q_i x_i + x_i^T c_i(t) + d_i(t), \\ \text{s.t.} \quad & \sum_{i=1}^m D_i x_i = b(t). \end{aligned} \quad (22)$$

In this problem, the Hessian matrix Q_i and the constraint matrix D_i for agent i are time-invariant.

Theorem 1. Under Assumptions 1-3, the i th agent's state vector $\hat{x}_i$ and Lagrange multiplier $\hat{\mu}_i$ of the dynamics (15) with (14) are convergent to the optimal trajectories and the Lagrange multiplier respectively for solving the optimization problem (22), if the matrices Q_i and D_i are time-invariant, and the parameter α satisfies (17).

Proof: If $Q_i(t)$ and $D_i(t)$ are time-invariant, we have $\dot{Q}_i(t)$ and $\dot{D}_i(t)$ are both zero matrices. Then from Lemma 5, if α satisfies the condition in (17), the state vector $\hat{\mu}_i$ of the distributed multi-agent system in (15) will reach consensus. That is $\lim_{t \rightarrow \infty} (\hat{\mu}_i(t) - \hat{\mu}_j(t)) = 0$ for $i, j = 1, 2, \dots, m$.

From Lemma 4, there exists a finite time $T > 0$ such that for any $t > T$ the equality (16) holds. Furthermore, due to $\dot{Q}_i(t) = 0$ and $\dot{D}_i(t) = 0$, (16) becomes

$$\frac{1}{m} \sum_{i=1}^m \dot{\hat{\mu}}_i - \dot{\mu} = -\frac{\sigma}{m} \sum_{i=1}^m (\hat{\mu}_i - \mu).$$

Then we get that $\sum_{i=1}^m (\hat{\mu}_i - \mu)$ is exponentially convergent to zero.

Combining with $\lim_{t \rightarrow \infty} (\hat{\mu}_i - \hat{\mu}_j) = 0$, it follows $\lim_{t \rightarrow \infty} (\hat{\mu}_i - \mu) = 0$; i.e., the state vector $\hat{\mu}_i(t)$ of the distributed multi-agent system in (15) will converge to $\mu(t)$.

In Lemma 3, combining with the expressions of w_i and $\hat{w}_i$, we have $x_i = Q_i^{-1}(t)(w_i - D_i^T(t)\mu)$ and $\hat{x}_i = Q_i^{-1}(t)(\hat{w}_i - D_i^T(t)\hat{\mu}_i)$. Then it follows $\lim_{t \rightarrow \infty} (\hat{x}_i - x_i) = 0$ due to $\lim_{t \rightarrow \infty} (\hat{w}_i - w_i) = 0$ and $\lim_{t \rightarrow \infty} (\hat{\mu}_i - \mu) = 0$.

Finally, since the state of the centralized multi-agent system (13) is convergent to the optimal trajectories $x_i^*(t)$ and $\mu^*(t)$ for solving the problem (3), that is, $\lim_{t \rightarrow \infty} (x_i(t) - x_i^*(t)) = 0$ and $\lim_{t \rightarrow \infty} (\mu(t) - \mu^*(t)) = 0$, the i th agent's state $\hat{x}_i(t)$ and the Lagrange multiplier $\hat{\mu}_i(t)$ in the distributed multi-agent system (15) with (14) are convergent to the corresponding optimal trajectories $x_i^*(t)$ and $\mu^*(t)$, respectively. ■

Remark 3. There are a broad spectrum of practical applications in the case where the Hessian matrices and constraint matrices are time-invariant and nonidentical. For example, it can be used to solve the target encirclement problem of multi-robot system shown in Section V-A, the economic dispatch problem in [36], and the time-varying distributed multi-energy management problem in [37]. It has been noted that the optimization problem (22) is also investigated in [14] for one-dimensional optimal resource allocation. However, here we investigate a more general problem (22) for high-dimensional case.

B. With Time-varying and Identical Hessian and Constraint Matrices

Now we are concerned with the time-varying quadratic programming as follows

$$\begin{aligned} \min_{x(t)} \quad & \sum_{i=1}^m \frac{1}{2} x_i^T Q(t) x_i + x_i^T c_i(t) + d_i(t), \\ \text{s.t.} \quad & \sum_{i=1}^m D(t) x_i = b(t). \end{aligned} \quad (23)$$

In this problem, the agents have identical time-varying Hessian and constraint matrices.

Theorem 2. Under Assumptions 1-3, the i th agent's state $\hat{x}_i$ and the Lagrange multiplier $\hat{\mu}_i$ of the multi-agent system in (15) with (14) are convergent to the corresponding optimal trajectories and the Lagrange multiplier respectively for solving the time-varying quadratic programming problem (23), if the matrices $Q_i(t)$ and $D_i(t)$ are identical for all agents, the conditions in (19) and (20) are satisfied, and σ satisfies

$$\sigma I_l + \frac{1}{2} (S(t) + S^T(t)) \succeq \eta I_l, \quad (24)$$

where $\eta > 0$ is a constant, and $S(t) = (D(t)Q^{-1}(t)D^T(t))^{-1} [D(t)Q^{-1}(t)(\dot{D}^T(t) - \dot{Q}(t)D^T(t)) + \dot{D}(t)D^T(t)]$.

Proof: Since the matrices $Q_i(t)$ and $D_i(t)$ (denoted as $Q(t)$ and $D(t)$ respectively) are identical for all agents, the matrix $\Lambda(t)$ in (13) becomes $\Lambda(t) = mD(t)Q^{-1}(t)D^T(t)$. Thus the system (14) does not need to estimate $\Lambda(t)$ for

this case. The matrix $m\Lambda_i^{-1}(t)$ in (15b) is replaced by $(D(t)Q^{-1}(t)D^T(t))^{-1}$.

According to Lemma 6, if the parameters σ and α satisfy the conditions in (19) and (20), the state vector $\hat{\mu}_i$ of the distributed multi-agent system in (15) will reach consensus. That is $\lim_{t \rightarrow \infty} (\hat{\mu}_i(t) - \hat{\mu}_j(t)) = 0$ ($i, j = 1, 2, \dots, m$).

By Lemma 4, there exists $T > 0$ such that for $t > T$ the equality (16) holds. Also, since the Hessian $Q_i(t)$ and constraint matrix $D_i(t)$ are identical for all agents, (16) becomes

$$\begin{aligned} & \frac{1}{m} \sum_{i=1}^m \dot{\hat{\mu}}_i - \dot{\mu} \\ &= -\frac{1}{m} (D(t)Q^{-1}(t)D^T(t))^{-1} (D(t)Q^{-1}(t)\dot{Q}(t) \\ & \quad - \dot{D}(t))Q^{-1}(t) \sum_{i=1}^m (\hat{w}_i - w_i) - \left\{ \frac{1}{m} (D(t)Q^{-1}(t)D^T(t))^{-1} \right. \\ & \quad \times \left[D(t)Q^{-1}(t)(\dot{D}^T(t) - \dot{Q}(t)D^T(t)) \right. \\ & \quad \left. \left. + \dot{D}(t)D^T(t) \right] + \frac{\sigma I_l}{m} \right\} \sum_{i=1}^m (\hat{\mu}_i - \mu). \end{aligned} \quad (25)$$

We denote $y = \sum_{i=1}^m (\hat{\mu}_i - \mu)$, $u = \sum_{i=1}^m (\hat{w}_i - w_i)$, $G(t) = (D(t)Q^{-1}(t)D^T(t))^{-1} (D(t)Q^{-1}(t)\dot{Q}(t) - \dot{D}(t))Q^{-1}(t)$, and $S(t) = (D(t)Q^{-1}(t)D^T(t))^{-1} [D(t)Q^{-1}(t)(\dot{D}^T(t) - \dot{Q}(t)D^T(t)) + \dot{D}(t)D^T(t)]$. Then, (25) is compactly written as

$$\dot{y} = -G(t)u - (S(t) + \sigma I_l)y. \quad (26)$$

From the proof of Lemma 3, we have $\dot{u} = -\sigma u$. Then it is straightforward to get $u(t) = u(0)e^{-\sigma t}$. We define a function $V(t) = y^T y / 2$. Then the time derivative of $V(t)$ along system (26) is provided by

$$\begin{aligned} \dot{V}(t) &= y^T \dot{y} \\ &= -y^T G(t)u - y^T (S(t) + \sigma I_l)y \\ &= -y^T G(t)u(0)e^{-\sigma t} - y^T \left(\frac{1}{2} (S(t) + S^T(t)) + \sigma I_l \right) y \\ &\leq \bar{G} \|y\|_2 \|u(0)\|_2 e^{-\sigma t} - \eta y^T y, \end{aligned} \quad (27)$$

where the condition (24) is used for the inequality derivation, and $\bar{G}$ is the uniform upper bound of $G(t)$ for $t \geq 0$; i.e., $\|G(t)\|_2 \leq \bar{G}$. Next, we will prove that $\|y(t)\|_2$ is exponentially convergent to zero.

In fact, if $\|y\|_2 \geq 2\bar{G}\|u(0)\|_2 e^{-\sigma t} / \eta$, one gets $\bar{G}\|u(0)\|_2 e^{-\sigma t} \leq \eta\|y\|_2 / 2$, and substituting it into (27), it follows $\dot{V}(t) \leq -\frac{\eta}{2} y^T y$.

Since $V(t) = y^T y / 2$, we get $V(t) \leq V(0)e^{-\eta t}$, which further results in $\|y\|_2 \leq \|y(0)\|_2 e^{-\eta t / 2}$. From above analysis, we get that $\|y\|_2$ always satisfies

$$\|y\|_2 \leq \max\{2\bar{G}\|u(0)\|_2 / \eta, \|y(0)\|_2\} e^{-\min\{\sigma, \eta/2\}t}.$$

Consequently, $\|y(t)\|_2$ is exponentially convergent to zero. Furthermore, because of $y = \sum_{i=1}^m (\hat{\mu}_i - \mu)$, it results in $\sum_{i=1}^m (\hat{\mu}_i - \mu)$ to be convergent to zero.

It is noted that the remainder of the proof is similar to that in Theorem 1, so we elide it here. ■

Remark 4. There are numerous practical applications in the case where the Hessian matrices and constraint matrices are time-varying and identical. For example, it can be used to solve the resource allocation problem, as demonstrated in [14] and [38]. It is noted that the optimization problem (23) is also investigated in [14] for addressing the one-dimensional optimal resource allocation problem with $D(t) = 1$. Consequently, the problem (23) investigated herein includes more general ones for this case.

V. SIMULATION RESULTS

In this section, the effectiveness of the designed distributed time-varying optimization algorithm is demonstrated by several numerical examples. In Section V-A, it is shown that the result of this paper can be employed for multi-robot system to collaboratively encircle a target. Then, the distributed time-varying quadratic programming with time-invariant and nonidentical Hessian and constraint matrices is verified by a numerical example in Section V-B. The distributed time-varying quadratic programming with time-varying and identical Hessian and constraint matrices is verified by a numerical example in Section V-C. Finally, numerical examples are provided in Section V-D to compare the proposed unified method (15) and algorithms in [14].

A. Application to target encirclement of multi-robot system

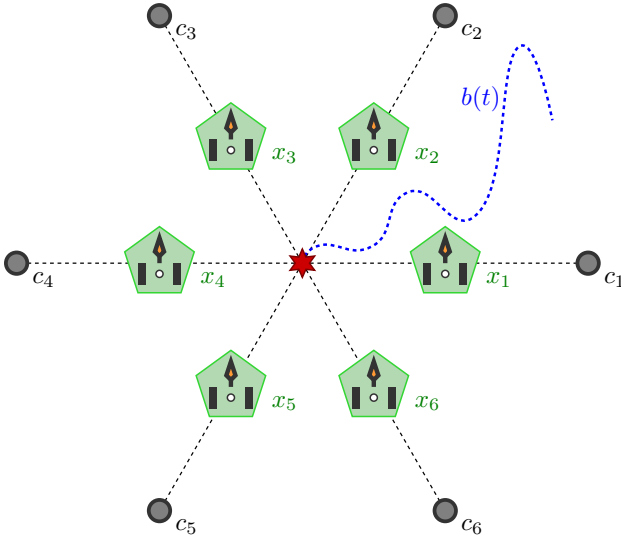


Fig. 1: Illustration of the target encirclement of a multi-agent system consisting of six robots in 2-D plane. The red six-pointed-star represents a moving target to be encircled, the six green robots are tasked to encircle the target, and the black points denote monitors.

In this part, the designed approach is applied to target encirclement of multi-robot system. In Fig. 1, it demonstrates a multi-robot system consisting of six robots, which is used to encircle a moving target cooperatively in 2-D plane. Assume that each monitor can only detect the corresponding robot and all monitors are always located at the circumference of a circle with the target as the centre and r as the radius. Let

$x_i = (x_i^1, x_i^2)^T \in \mathbb{R}^2$ denotes the state of robot i , specially the location of robot i in this example, $b = (b^1, b^2)^T \in \mathbb{R}^2$ shows the state of the moving target, and $c_i = (c_i^1, c_i^2)^T \in \mathbb{R}^2$ represents the state of monitor i .

To encircle the target $b(t)$, the center of all robots is required to be consistent with the target each time. That is, the following one condition should be held

$$\frac{1}{6} \sum_{i=1}^6 x_i(t) = b(t) \quad (28)$$

To avoid colliding with the target and being detected by the corresponding monitor c_i , the robot i is required to be as close to the monitor c_i as possible. In addition, the target should be detected by the robots. What is more, because the sensing ability of each monitor to the corresponding robot may differ in practical applications, the weight parameter λ_i is set in the i -th local objective function for $\lambda_i \in [0, 1]$ ($i = 1, 2, \dots, 6$). Based on the above analysis, the local objective function for the i th robot can be formulated as $f_i(x_i, t) = (1 - \lambda_i)\|x_i - b(t)\|_2^2/2 + \lambda_i\|x_i - c_i(t)\|_2^2/2$. For better performance, all robots work collaboratively to minimize the total cost function $f(x, t) = \sum_{i=1}^6 f_i(x_i, t)$. In this way, the robots are able to reduce the cost by ensuring that each robot is contributing optimally to the system.

Combining with the global equality constraint (28), the target encirclement problem of multi-robot system is specifically expressed as

$$\begin{aligned} \min_x \quad & \frac{1}{2} \sum_{i=1}^6 (1 - \lambda_i) \|x_i - b(t)\|_2^2 + \lambda_i \|x_i - c_i(t)\|_2^2, \\ \text{s. t.} \quad & \frac{1}{6} \sum_{i=1}^6 x_i(t) = b(t), \end{aligned}$$

where $b(t) = \sum_{i=1}^6 b_i(t) = \sum_{i=1}^6 (t, i \cos(0.5it)/3)^T$ denotes the time-varying position of the target at time t .

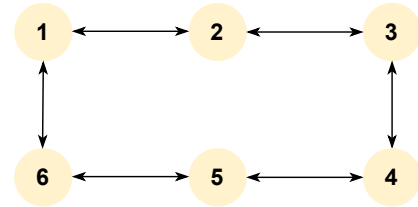


Fig. 2: Communication graph among six agents.

In this case, as seen in Fig. 2, the agents communicate with their neighbors through a network with undirected and connected graph. And the parameters in this problem are set as: $(\lambda_1, \lambda_2, \dots, \lambda_6) = (0.3, 0.4, 0.5, 0.7, 0.6, 0.5)$, $\sigma = 3$, $\alpha = 1000$, and $r = 10$. The initial positions of the six robots are placed randomly within the region $[-5, 5]^2$.

The trajectories of all robots and the moving target are presented in Fig. 3. The red triangles are the initial positions of robots and the blue asterisks are the end positions of all robots at $t = 4s$. Furthermore, the bright blue, orange and green points stand for the states of all robots and the target at $t = 1s$, $t = 2s$ and $t = 3s$, respectively. It is noticeable that the robots can track the corresponding optimal trajectory and

encircle the moving target after a specific time. Additionally, the convergent behavior of the relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ is depicted in Fig. 4.

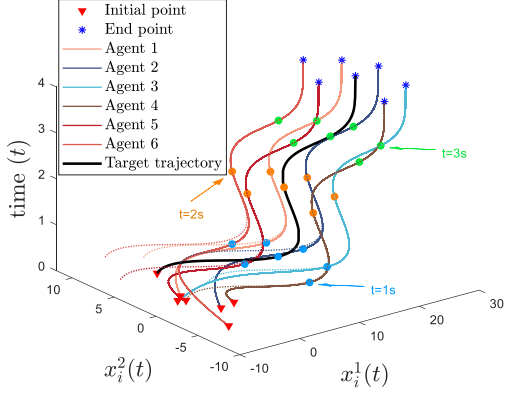


Fig. 3: Trajectories of robots under distributed multi-agent system (15) as well as the trajectory of moving target within $t = 4$ s. Dashed lines denote the optimal trajectories of robots.

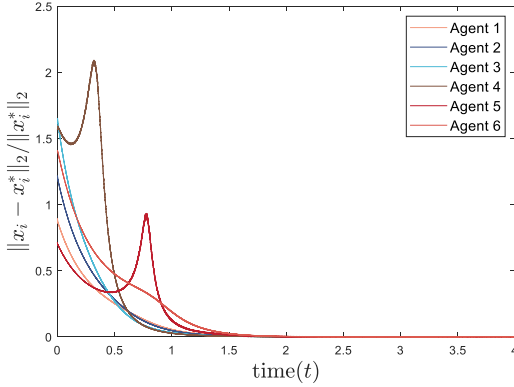


Fig. 4: The relative error between the solution obtained by the proposed multi-agent system (15) and the corresponding optimal solution, which is given as $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ for the example in Section V-A.

B. Simulation with time-invariant and nonidentical Hessian and constraint matrices

In order to verify the result in Theorem 1 for resolving time-varying quadratic programming with time-invariant and nonidentical Hessian and constraint matrices, we give a numerical example with $m = 4$, $n = 2$ and $l = 2$. The topology graph is a ring structure similar to Fig. 2. Let $x_i = (x_i^1, x_i^2)^T \in \mathbb{R}^2$ and $b = (b^1, b^2)^T \in \mathbb{R}^2$. In the problem (22), we specify the terms as follows: $b(t) = \sum_{i=1}^m b_i(t)$, $b_i(t) = (it, 2e^{-t/i} + i)^T$, $c_i(t) = (i, e^{-it})^T$, $d_i(t) = 0$, $D_i = iI_2$, $Q_i = \begin{pmatrix} i^2 + 2e^{-0.1i} & \\ e^{-0.1i} & i^2 \end{pmatrix}$.

By simple calculations, the parameters in the proposed multi-agent system (15) are given as follows: $\sigma = 5$, $\alpha = 500$. The initial states for all agents are selected randomly within a 2-dimensional region $[-5, 5]^2$, and are shown as red triangles in Fig. 5. The trajectories of agents are given in Fig. 5 too, and at time $t = 4$ s, the blue asterisks represent the end positions

of agents. Fig. 5 shows that the agents' states are convergent to the corresponding optimal trajectories. Moreover, the convergent behavior of the relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ is depicted in Fig. 6.

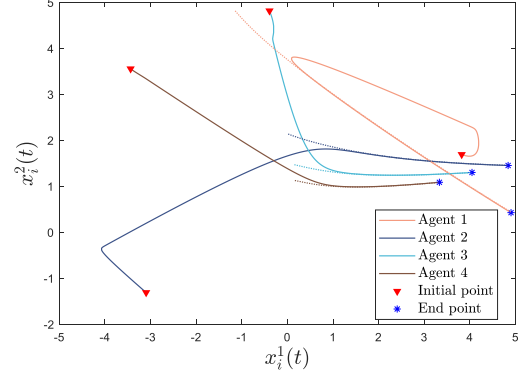


Fig. 5: Trajectories of agents within $t = 4$ s under distributed multi-agent system (15). The red triangles denote initial points of all agents, while the blue asterisks denote the end points. Dashed lines denote the optimal trajectories of agents.

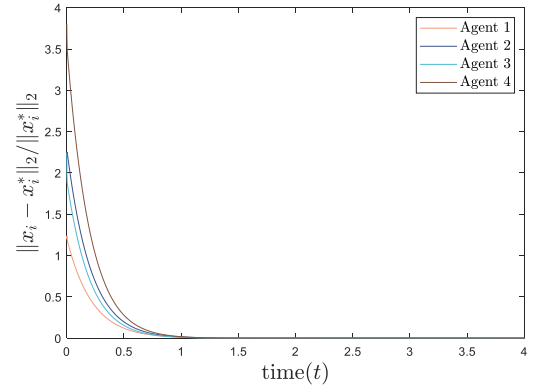


Fig. 6: The relative error between the solution obtained by the proposed multi-agent system (15) and the corresponding optimal solution, which is given as $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ for the example in Section V-B.

C. Simulation with time-varying and identical Hessian and constraint matrices

To verify the result in Theorem 2 for resolving time-varying quadratic programming with time-varying and identical Hessian and constraint matrices, we give a numerical example with $m = 6$, $n = 3$ and $l = 2$. The topology graph is as the same as that in Fig. 2. Let $x_i = (x_i^1, x_i^2, x_i^3)^T \in \mathbb{R}^3$ and $b = (b^1, b^2)^T \in \mathbb{R}^2$. In the problem (23), we specify the terms as follows: $b(t) = \sum_{i=1}^m b_i(t)$, $b_i(t) = (5it, it^2)^T$, $c_i(t) = (2, 3e^{-it}, e^{-it})^T$, $d_i(t) = 0$,

$$Q(t) = \begin{pmatrix} -3e^{-0.1t} + 4 & 1.5 \sin(t) & \cos(t) \\ 1.5 \sin(t) & 0.5e^{-2t} + 1 & 1 \\ \cos(t) & 1 & e^{-t} + 2 \end{pmatrix},$$

$$D(t) = \begin{pmatrix} e^{-t} & 0 & 1 - e^{-t} \\ 0 & e^{-0.4t} & 1 - e^{-0.4t} \end{pmatrix}.$$

By simple calculations, the parameters in the proposed multi-agent system (15) are given as follows: $\sigma = 4$, $\alpha = 600$. The initial states for all agents are selected randomly within a 3-dimensional region $[-50, 50]^3$, and are shown as red triangles in Fig. 7. The trajectories of agents are given in Fig. 7 too, and at time $t = 3$ s, the blue asterisks represent the end positions of agents. Fig. 7 shows that the agents' states are convergent to the corresponding optimal trajectories. Moreover, the convergent behavior of the relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ is depicted in Fig. 8.

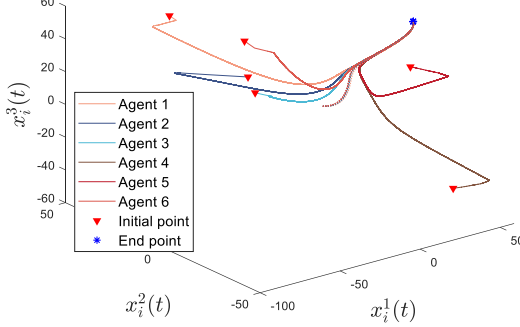


Fig. 7: Trajectories of agents within $t = 3$ s under distributed multi-agent system (15). The red triangles denote initial points of all agents, while the blue asterisks denote the end points. Dashed lines denote the optimal trajectories of agents.

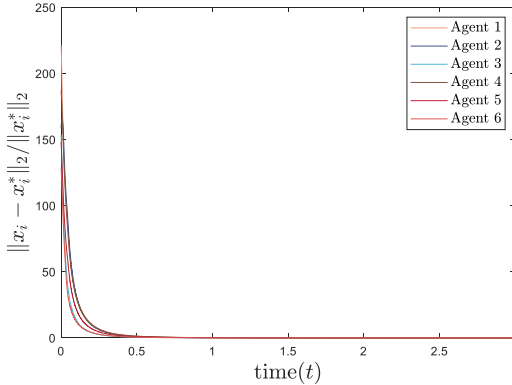


Fig. 8: The relative error between the solution obtained by the proposed multi-agent system (15) and the corresponding optimal solution, which is given as $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ for the example in Section V-C.

D. Comparisons with algorithms in [14]

Because the problem investigated in this paper is more general, the algorithms in [14] cannot be applied to the above three simulation examples. In this subsection, numerical simulations with $m = 6$, $n = 1$ and $l = 1$ are provided to make comparisons between the proposed unified method (15) and algorithms in [14]. In scenario 1, the terms in (22) are specified as: $Q_1 = 3.1211$, $Q_2 = 3.5716$, $Q_3 = 4.5144$, $Q_4 = 4.9527$, $Q_5 = 3.7608$, $Q_6 = 4.1356$, $c_i(t) = -2ia_i \sin(\frac{\pi}{4}t)$, $d_i(t) = i^2 \sin^2(\frac{\pi}{4}t)$, $b(t) = b = 120$, $b_i = b/m$, and $D_i = 1$. The parameters in (15) are set as: $\sigma = 3$, $\alpha = 50$. The parameters in (30) and (31) of [14] are set as: $\alpha = 3$, $\eta_0 = 50$,

$\beta = 0.01$, and $\gamma = 1$. In scenario 2, the terms in (23) are specified as: $Q(t) = 8 + 2\sin(0.34t)$, $c_i(t) = -4i\cos(0.5it)$, $d_i(t) = i^2 \sin(0.5it)$, $b(t) = b = 120$, $b_i = b/m$, and $D(t) = 1$. The parameters in (15) are set as: $\sigma = 3$, $\alpha = 100$. The parameters in (19) of [14] are set as: $\alpha = 3$, $\tau = 100$.

The results of scenario 1 are presented in Fig. 9 and the results of scenario 2 are presented in Fig. 10. As shown in Fig. 9 and Fig. 10, each agent can track the corresponding optimal solution under the proposed unified algorithm and algorithms in [14] for the two scenarios. The performance of the two algorithms in Fig. 10 is very similar because the algorithm proposed in this paper can be simplified to the algorithm in [14] under the problem defined in scenario 2.

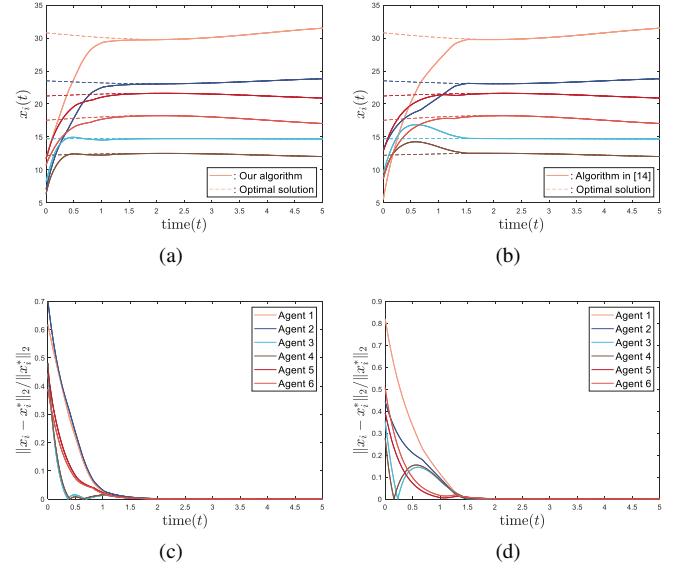


Fig. 9: Results of scenario 1 with time-invariant and non-identical Hessians: (a) The agents' states under (15); (b) The agents' states under (30) and (31) in [14]; (c) The relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ under (15); (d) The relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ under (30) and (31) in [14].

VI. CONCLUSIONS

In this paper, a unified distributed multi-agent system has been presented for constrained time-varying quadratic programming in two distinct cases: one is that the optimization problem has time-invariant and nonidentical Hessian and constraint matrices, the other one is that it has time-varying and identical Hessian and constraint matrices. We began with developing a centralized algorithm for the time-varying constrained quadratic programming using the prediction-correction method. Then, a multi-agent system was proposed using the distributed average tracking method to estimate some global information. The states of agents obtained by the proposed multi-agent system have been certified to accurately track the corresponding optimal trajectories for a wide range of parameters, and even in the presence of some mild conditions. In addition, the proposed approach has been shown to be effective in enabling multiple robots to achieve their goal of encircling a target.

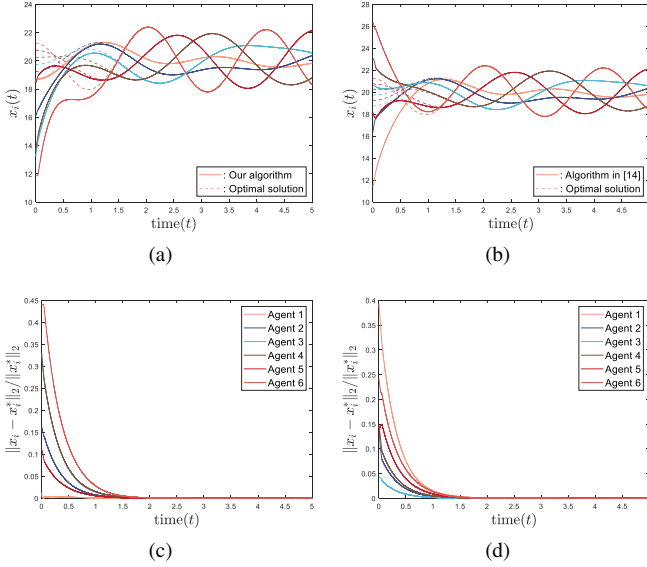


Fig. 10: Results of scenario 2 with time-varying and identical Hessians: (a) The agents' states under (15); (b) The agents' states under (19) in [14]; (c) The relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ under (15); (d) The relative error $\|x_i - x_i^*\|_2 / \|x_i^*\|_2$ under (19) in [14].

APPENDIX

A. Proof of Lemma 2

The proof is an extension of Lemma 1 in [14], which generalizes the scalar system to vector system. In a convenient way, we denote $\bar{z}(t) = (\sum_{j=1}^m z_j(t))/m$ and $e_i(t) = z_i(t) - \bar{z}(t)$. From symmetry of the graph in Assumption 1, we get $\dot{\bar{z}}(t) = \frac{1}{m} \sum_{j=1}^m \dot{z}_j(t) = \frac{1}{m} \sum_{j=1}^m (-\Xi(t)z_j(t) + v_j(t))$.

Then it follows $\dot{z}_i(t) - \dot{\bar{z}}(t) = -\Xi(t)z_i(t) + \frac{1}{m} \sum_{j=1}^m \Xi(t)z_j(t) + v_i(t) - \frac{1}{m} \sum_{j=1}^m v_j(t) - \alpha \sum_{j \in N_i} \text{sign}(z_i(t) - z_j(t))$.

To get the consensus of the system, we take $V(t) = (1/2) \sum_{i=1}^m e_i^T(t)e_i(t)$ as the Lyapunov function candidate. Then the time derivative of $V(t)$ along the dynamical system (2) can be derived

$$\begin{aligned} \dot{V}(t) &= \sum_{i=1}^m e_i^T(t) \dot{e}_i(t) = \sum_{i=1}^m e_i^T(t) (\dot{z}_i(t) - \dot{\bar{z}}(t)) \\ &= \sum_{i=1}^m e_i^T(t) \left(-\Xi(t)(z_i(t) - \frac{1}{m} \sum_{j=1}^m z_j(t)) \right. \\ &\quad \left. + v_i(t) - \frac{1}{m} \sum_{j=1}^m v_j(t) - \alpha \sum_{j \in N_i} \text{sign}(z_i(t) - z_j(t)) \right) \\ &= \sum_{i=1}^m e_i^T(t) \left(-\Xi(t)e_i(t) + v_i(t) \right. \\ &\quad \left. - \alpha \sum_{j \in N_i} \text{sign}(z_i(t) - z_j(t)) \right). \end{aligned} \quad (29)$$

where the last equality works due to $e_i = z_i(t) - (\sum_{j=1}^m z_j(t))/m$, and it follows $\sum_{i=1}^m e_i^T(t) \sum_{j=1}^m v_j(t) = 0$.

Under Assumption 1, we get

$$\begin{aligned} &\sum_{i=1}^m e_i^T(t) \sum_{j \in N_i} \text{sign}(z_i(t) - z_j(t)) \\ &= \frac{1}{2} \sum_{i=1}^m \sum_{j \in N_i} (e_i(t) - e_j(t))^T \text{sign}(z_i(t) - z_j(t)) \\ &= \frac{1}{2} \sum_{i=1}^m \sum_{j \in N_i} (e_i(t) - e_j(t))^T \text{sign}(e_i(t) - e_j(t)) \\ &= \frac{1}{2} \sum_{i=1}^m \sum_{j \in N_i} \|e_i(t) - e_j(t)\|_1, \end{aligned} \quad (30)$$

in which the first equality is from Lemma 6.1 in [39], and the second equality is ensured by the definition of $e_i(t)$.

Substituting (30) into (29) results in

$$\begin{aligned} \dot{V}(t) &= - \sum_{i=1}^m e_i^T(t) \Xi(t) e_i(t) + \sum_{i=1}^m e_i^T(t) v_i(t) \\ &\quad - \frac{\alpha}{2} \sum_{i=1}^m \sum_{j \in N_i} \|e_i(t) - e_j(t)\|_1 \\ &\leq - \frac{1}{2} \sum_{i=1}^m e_i^T(t) (\Xi(t) + \Xi^T(t)) e_i(t) \\ &\quad + \sum_{i=1}^m \|e_i(t)\|_1 \|v_i(t)\|_\infty - \frac{\alpha}{2} \sum_{i=1}^m \sum_{j \in N_i} \|e_i(t) - e_j(t)\|_1 \\ &\leq -\tau \sum_{i=1}^m \|e_i(t)\|_2^2 + \kappa \sum_{i=1}^m \|e_i(t)\|_1 \\ &\quad - \frac{\alpha}{2} \sum_{i=1}^m \sum_{j \in N_i} \|e_i(t) - e_j(t)\|_1, \end{aligned} \quad (31)$$

in which the second inequality holds because $(\Xi(t) + \Xi^T(t))/2 \succeq \tau I_n$ and $\sup_{t \geq 0} \|v_i(t)\|_\infty < \kappa$.

According to the properties of norms, we have $\sum_{i=1}^m \|e_i(t)\|_1 \leq \sqrt{nm} (\sum_{i=1}^m \|e_i(t)\|_2^2)^{1/2}$ and $\|e_i(t) - e_j(t)\|_1 \geq \|e_i(t) - e_j(t)\|_2$. Then (31) derives

$$\begin{aligned} \dot{V}(t) &\leq -\tau \sum_{i=1}^m \|e_i(t)\|_2^2 + \kappa \sqrt{nm} \left(\sum_{i=1}^m \|e_i(t)\|_2^2 \right)^{\frac{1}{2}} \\ &\quad - \frac{\alpha}{2} \sum_{i=1}^m \sum_{j \in N_i} \|e_i(t) - e_j(t)\|_2 \\ &\leq -\tau \sum_{i=1}^m \|e_i(t)\|_2^2 + \kappa \sqrt{nm} \left(\sum_{i=1}^m \|e_i(t)\|_2^2 \right)^{\frac{1}{2}} \\ &\quad - \frac{\alpha}{2} \left(\sum_{i=1}^m \sum_{j \in N_i} \|e_i(t) - e_j(t)\|_2^2 \right)^{\frac{1}{2}} \\ &\leq -\tau \sum_{i=1}^m \|e_i(t)\|_2^2 + \kappa \sqrt{nm} (2V(t))^{\frac{1}{2}} \\ &\quad - \frac{\alpha}{2} \left(2\lambda_2(L) \sum_{i=1}^m \|e_i(t)\|_2^2 \right)^{\frac{1}{2}} \\ &= -2\tau V(t) - \left(\alpha \sqrt{\lambda_2(L)} - \kappa \sqrt{2nm} \right) V^{\frac{1}{2}}. \end{aligned}$$

Then if $\alpha \geq \kappa \sqrt{2nm/\lambda_2(L)}$, we get $V(t)$ to be convergent to zero exponentially. Thus the state vector $z_i(t)$ of multi-agent system (2) will reach consensus.

B. Proof of Lemma 3

Following the definition of w_i , one gets $\dot{w}_i = \dot{Q}_i(t)x_i + Q_i(t)\dot{x}_i + \dot{D}_i^T(t)\mu + D_i^T(t)\dot{\mu}$. Substituting (13) into the previous equality, one gets $\dot{w}_i = \dot{Q}_i(t)x_i - \sigma \nabla_{x_i} f_i(x_i, t) - \sigma D_i^T(t)\mu - \nabla_{x_i} f_i(x_i, t)$.

Due to $\nabla_{x_i} f_i(x_i, t) = Q_i(t)x_i + c_i(t)$ and $\nabla_{x_i} f_i(x_i, t) = \dot{Q}_i(t)x_i + \dot{c}_i(t)$, it follows

$$\dot{w}_i = -\sigma w_i - \sigma c_i(t) - \dot{c}_i(t). \quad (32)$$

In the same way, it is similarly to get

$$\dot{\hat{w}}_i = -\sigma \hat{w}_i - \sigma c_i(t) - \dot{c}_i(t). \quad (33)$$

Subtracting both sides of the equations in (32) and (33), we get $\dot{\hat{w}}_i - \dot{w}_i = -\sigma(\hat{w}_i - w_i)$, which implies $\lim_{t \rightarrow \infty}(\hat{w}_i - w_i) = 0$.

C. Proof of Lemma 4

With considering $\nabla_{x_i} f_i(x_i, t) = Q_i(t)x_i + c_i(t)$ and $\nabla_{x_i} f_i(x_i, t) = \dot{Q}_i(t)x_i + \dot{c}_i(t)$, from (15b), one gets

$$\begin{aligned} \frac{1}{m} \sum_{i=1}^m \dot{\hat{\mu}}_i &= \sum_{i=1}^m \Lambda_i^{-1}(t) \left[-D_i(t)Q_i^{-1}(t)(\sigma \nabla_{x_i} f_i(\hat{x}_i, t) \right. \\ &\quad \left. + \nabla_{x_i} f_i(\hat{x}_i, t) + \dot{D}_i^T(t)\hat{\mu}_i) + \sigma(D_i(t)\hat{x}_i - b_i(t)) \right. \\ &\quad \left. + (\dot{D}_i(t)\hat{x}_i - \dot{b}_i(t)) \right] - \frac{\sigma}{m} \sum_{i=1}^m \hat{\mu}_i \\ &= -\sum_{i=1}^m \Lambda_i^{-1}(t)(D_i(t)Q_i^{-1}(t)\dot{Q}_i(t) - \dot{D}_i(t))\hat{x}_i \\ &\quad - \sum_{i=1}^m \left(\Lambda_i^{-1}(t)D_i(t)Q_i^{-1}(t)\dot{D}_i^T(t) + \frac{\sigma I_l}{m} \right) \hat{\mu}_i \\ &\quad - \sum_{i=1}^m \Lambda_i^{-1}(t) \left[D_i(t)Q_i^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) \right. \\ &\quad \left. + \sigma b_i(t) + \dot{b}_i(t) \right], \end{aligned} \quad (34)$$

and from (13b), it follows

$$\begin{aligned} \dot{\mu} &= -\Lambda^{-1}(t) \sum_{i=1}^m (D_i(t)Q_i^{-1}(t)\dot{Q}_i(t) - \dot{D}_i(t))x_i \\ &\quad - \sum_{i=1}^m \left(\Lambda^{-1}(t)D_i(t)Q_i^{-1}(t)\dot{D}_i^T(t) + \frac{\sigma I_l}{m} \right) \mu \\ &\quad - \Lambda^{-1}(t) \sum_{i=1}^m \left[D_i(t)Q_i^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) \right. \\ &\quad \left. + \sigma b_i(t) + \dot{b}_i(t) \right], \end{aligned} \quad (35)$$

in which the condition $b(t) = \sum_{i=1}^m b_i(t)$ is used.

According to the definition of $\theta_i(t)$ and Assumptions 2-3, $\sup_{t \geq 0} \|\theta_i(t)\|_\infty$ is bounded. For system (14), according to Lemma 1, the output vector $z_i(t)$ will reach consensus in finite time provided that the design parameter γ satisfies $\gamma >$

$\bar{\theta} \geq \sup_{t \geq 0} \|\dot{\theta}_i(t)\|_\infty$; i.e., there exists a finite time $T > 0$ with $T \leq (\gamma - \bar{\theta})/2 \sum_{i=1}^m \sum_{j \in N_i} \|z_i(0) - z_j(0)\|_\infty$, and we have $z_i(t) = (\sum_{i=1}^m \theta_i(t))/m$ for $t \geq T$. Furthermore, from the definition of $\Lambda_i(t)$, it results in $\Lambda_i(t) = \Lambda(t)$ for $t \geq T$.

From the equalities in (34) and (35), when $t \geq T$, we get

$$\begin{aligned} &\frac{1}{m} \sum_{i=1}^m \dot{\hat{\mu}}_i - \dot{\mu} \\ &= -\Lambda^{-1}(t) \sum_{i=1}^m (D_i(t)Q_i^{-1}(t)\dot{Q}_i(t) - \dot{D}_i(t))(\hat{x}_i - x_i) \\ &\quad - \sum_{i=1}^m \left(\Lambda^{-1}(t)D_i(t)Q_i^{-1}(t)\dot{D}_i^T(t) + \frac{\sigma I_l}{m} \right) (\hat{\mu}_i - \mu). \end{aligned} \quad (36)$$

Then from the expressions of w_i and $\hat{w}_i$, it follows $x_i = Q_i^{-1}(t)(w_i - D_i^T(t)\mu)$ and $\hat{x}_i = Q_i^{-1}(t)(\hat{w}_i - D_i^T(t)\hat{\mu}_i)$. Then $\hat{x}_i - x_i = Q_i^{-1}(t)(\hat{w}_i - w_i) - D_i^T(t)(\hat{\mu}_i - \mu)$ and substituting it into (36) results in the equality (16).

D. Proof of Lemma 5

According to (15b), with considering $\nabla_{x_i} f_i(x_i, t) = Q_i(t)x_i + c_i(t)$ and $\nabla_{x_i} f_i(x_i, t) = \dot{Q}_i(t)x_i + \dot{c}_i(t)$, $\dot{\mu}_i$ is further written as $\dot{\mu}_i = -\sigma \hat{\mu}_i - m\Lambda_i^{-1}(t)(D_i(t)Q_i^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) + \sigma b_i(t) + \dot{b}_i(t)) - \alpha \sum_{j \in N_i} \text{sign}(\hat{\mu}_i - \hat{\mu}_j)$, which is simplified as $\dot{\mu}_i = -\sigma \hat{\mu}_i + \hat{q}_i(t) - \alpha \sum_{j \in N_i} \text{sign}(\hat{\mu}_i - \hat{\mu}_j)$, where $\hat{q}_i(t) = -m\Lambda_i^{-1}(t)(D_i(t)Q_i^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) + \sigma b_i(t) + \dot{b}_i(t))$.

As per Assumptions 2 and 3, $\hat{q}_i(t)$ is bounded by κ as estimated in (18). Then the conditions in Lemma 2 are satisfied if (17) holds, which follows that for $i = 1, 2, \dots, m$, $\hat{\mu}_i$ will reach consensus; i.e., $\lim_{t \rightarrow \infty}(\hat{\mu}_i - \hat{\mu}_j) = 0$.

E. Proof of Lemma 6

From (15b), with considering $\nabla_{x_i} f_i(x_i, t) = Q_i(t)x_i + c_i(t)$ and $\nabla_{x_i} f_i(x_i, t) = \dot{Q}_i(t)x_i + \dot{c}_i(t)$ and $\dot{Q}(t)x_i + \dot{c}_i(t)$, $\dot{\mu}_i$ is further written as

$$\begin{aligned} \dot{\mu}_i &= -(m\Lambda^{-1}(t)D(t)Q^{-1}(t)\dot{D}^T(t) + \sigma I_l)\hat{\mu}_i \\ &\quad + m\Lambda^{-1}(t)(\dot{D}(t) - D(t)Q^{-1}(t)\dot{Q}(t))\hat{x}_i \\ &\quad - m\Lambda^{-1}(t)(D(t)Q^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) \\ &\quad + \sigma b_i(t) + \dot{b}_i(t)) - \alpha \sum_{j \in N_i} \text{sign}(\hat{\mu}_i - \hat{\mu}_j). \end{aligned} \quad (37)$$

With considering the definition of $\hat{w}_i$, we have $\hat{x}_i = Q_i^{-1}(t)(\hat{w}_i - D_i^T(t)\hat{\mu}_i) = Q^{-1}(t)(\hat{w}_i - D^T(t)\hat{\mu}_i)$ and substituting it into (37) yields

$$\begin{aligned} \dot{\hat{\mu}}_i &= -\left\{ m\Lambda^{-1}(t) \left[(\dot{D}(t) - D(t)Q^{-1}(t)\dot{Q}^T(t))Q^{-1}(t)D^T(t) \right. \right. \\ &\quad \left. \left. + D(t)Q^{-1}(t)\dot{D}^T(t) \right] + \sigma I_l \right\} \hat{\mu}_i \\ &\quad + m\Lambda^{-1}(t)(\dot{D}(t) - D(t)Q^{-1}(t)\dot{Q}(t))Q^{-1}(t)\hat{w}_i \\ &\quad - m\Lambda^{-1}(t)(D(t)Q^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) \\ &\quad + \sigma b_i(t) + \dot{b}_i(t)) - \alpha \sum_{j \in N_i} \text{sign}(\hat{\mu}_i - \hat{\mu}_j), \end{aligned}$$

which is simplified as $\dot{\hat{\mu}}_i = -\delta(t)\hat{\mu}_i + q_i(t) - \alpha \sum_{j \in N_i} \text{sign}(\hat{\mu}_i - \hat{\mu}_j)$, where $\delta(t) = m\Lambda^{-1}(t)[(\dot{D}(t) -$

$D(t)Q^{-1}(t)\dot{Q}^T(t))Q^{-1}(t)D^T(t)+D(t)Q^{-1}(t)\dot{D}^T(t)]+\sigma I_L$, and $q_i(t) = m\Lambda^{-1}(t)[(\dot{D}(t)-D(t)Q^{-1}(t)\dot{Q}(t))Q^{-1}(t)\hat{w}_i - D(t)Q^{-1}(t)(\sigma c_i(t) + \dot{c}_i(t)) - \sigma b_i(t) - \dot{b}_i(t)]$.

From (33), it follows that $\hat{w}_i(t)$ is bounded with $\|\hat{w}_i(t)\|_\infty \leq \sup_{t \geq 0} \|c_i(t) + \dot{c}_i(t)/\sigma\|_\infty + \|\hat{w}_i(0)\|_\infty = \sup_{t \geq 0} \|c_i(t) + \dot{c}_i(t)/\sigma\|_\infty + \|Q(0)\hat{x}_i(0) + D^T(0)\hat{\mu}_i(0)\|_\infty$, and thus $q_i(t)$ is also bounded by κ as estimated in (21). If the condition in (19) is satisfied, we have $\delta_i(t) + \delta_i^T(t) \geq 2\tau I_L$. Then the conditions in Lemma 2 are satisfied if (19) and (20) hold, which follows that for $i = 1, 2, \dots, m$, $\hat{\mu}_i$ can reach consensus as $t \rightarrow \infty$; i.e., $\lim_{t \rightarrow \infty} (\hat{\mu}_i - \hat{\mu}_j) = 0$.

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